FAILURE OF CONVERGENCE TO MORSE GLOBAL MINIMA BELOW THE LOWER HESSIAN DRIFT THRESHOLD

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Editorial note. Only the low-drift nonconvergence Theorem theo2 is counted. The retained high-drift result and comparison construction are uncounted source context.

Allowed standard prerequisites (editorial).

- Ito formula, time changes of continuous local martingales, one-dimensional diffusion comparison and Bessel recurrence/transience. Standard stochastic-analysis facts cited in the original article; local Hessian comparison remains part of the supplied proof.

More precisely, on a compact Riemannian manifold M, of dimension $m \geq 1$, let U be a Morse function satisfying $\min_M U = 0$. Recall that a smooth mapping is a Morse function if its Hessian is non-degenerate at each of its critical points (which are the points where the gradient vanishes).

Consider a diffusion $X := (X(t))_{t \geq 0}$ associated to the generator

$$L_\beta := U\Delta \cdot -\beta\langle \nabla U, \nabla \cdot \rangle$$

where Δ, $\langle \cdot, \cdot \rangle$ and ∇ stand for the Laplacian, scalar product and gradient coming from the Riemannian structure, and where β is a real number. Since we want to find the global minima of U, we are more interested in the case $\beta > 0$, where the drift has an attractive (respectively repulsive) effect with respect to the local minima (resp. maxima). But as we are to see, it is convenient to also consider non-positive values of β, where the effects of the drift are reversed.

Due to the Morse assumption on U and the fact that $\min_M U = 0$, we have that $\sqrt{U}$ is a Lipschitz mapping on M (see Remark 1.1 below). As a consequence it is possible to construct X, whatever the initial condition, as the unique strong solution to a stochastic differential equation driven by a m-dimensional Brownian motion $B := (B(t))_{t \geq 0}$ (independent from $X(0)$), see for instance Ikeda and Watanabe [IW89]. Heuristically, this stochastic differential equation can be written under the Itô's form

$$dX(t) = -\beta \nabla U(X(t))\, dt + \sqrt{2U(X(t))}\, dB(t) \tag{1}$$

or under Stratonovich's form

$$dX(t) = -\left(\beta + \frac{1}{2\sqrt{2}}\right) \nabla U(X(t))\, dt + \sqrt{2U(X(t))} \circ dB(t)$$

where $dB(t)$ has to be isometrically interpreted in the tangent space above $X(t)$ through stochastic parallel displacements. At least when M is a flat torus, the writing (1) is perfectly rigorous.

Consider the set of global minima of U given by

$$\mathcal{U} := \{x \in M \,:\, U(x) = 0\}.$$

Note that if $X(0) \in \mathcal{U}$, then X does not move: for any $t \geq 0$, we have $X(t) = X(0)$. This observation can be strengthened into the attractiveness of $\mathcal{U}$, which is the main goal of this paper.

We need some additional notations. Under the Morse assumption, $\mathcal{U}$ consists of a finite set of points, say $y_1, y_2, \ldots, y_N$, with $N \in \mathbb{N}$. For each $n \in [\![N]\!] := \{1, 2, \ldots, N\}$, denote $\lambda_1(n) \leq \lambda_2(n) \leq \cdots \leq \lambda_m(n)$ the positive eigenvalues of the Hessian of U at y_n. Introduce the condition

$$\beta > \max_{n \in [\![N]\!]} \frac{\sum_{i \in [\![m]\!]} \lambda_i(n)}{2\lambda_1(n)} - 1. \tag{2}$$

Theorem 0.1. *Whatever the initial condition $X(0)$, under (2) the limit $X(\infty) := \lim_{t \to +\infty} X(t)$ exists a.s. and belongs to $\mathcal{U}$. Furthermore, when $m \geq 2$, if X starts from a point $x_0 \notin \mathcal{U}$, we have for any $y \in \mathcal{U}$,*

$$\mathbb{P}_{x_0}[X(\infty) = y] > 0.$$

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When $m = 1$, denote y_1 and y_2 the boundary points of the connected component of $M \setminus \mathcal{U}$ containing x_0 (when $\mathcal{U}$ is a singleton, we get $y_1 = y_2$). Then we have

$$\mathbb{P}_{x_0}[X(\infty) = y_1] \ > \ 0, \quad \mathbb{P}_{x_0}[X(\infty) = y_2] \ > \ 0,$$
$$\forall \ \ y \in \mathcal{U} \setminus \{y_1, y_2\}, \quad \mathbb{P}_{x_0}[X(\infty) = y] = 0.$$

Thus under (2) X is a time-homogeneous stochastic algorithm minimizing globally U and finding all the global minima as soon as $m \geq 2$. A.s. convergence was not considered in the previously cited recent works in Machine Learning, which rather concentrated on approximation properties (from discrete time stochastic gradient descent algorithms), on convergence in law and in the behaviour of the associated invariant measures.

As a consequence, the value $m/2 - 1$ is critical for β when at each global minimum the Hessian is proportional to the identity:

Corollary 0.2. *Assume that for each $n \in \mathbb{N}$ we have $\lambda_1(n) = \lambda_m(n)$, then we have*

$$\beta > \frac{m}{2} - 1 \Longrightarrow \mathbb{P}\left[\lim_{t \to +\infty} X(t) \text{ exists and belongs to } \mathcal{U}\right] = 1$$
$$\beta < \frac{m}{2} - 1 \Longrightarrow \mathbb{P}\left[\lim_{t \to +\infty} X(t) \text{ exists and belongs to } \mathcal{U}\right] = 0.$$

In particular this result always applies in dimension 1 and we get $-1/2$ as the critical value for β (justifying our consideration of negative β).

In practice we may not have access to the eigenvalues of the Hessian at the global minima, so Condition (2) is difficult to check. An alternative to the appropriate choice of the coefficient β is to let it depend on the current value of U. More precisely, consider $\zeta : (0, +\infty) \to (0, +\infty)$ a smooth function such that

$$\lim_{u \to 0_+} \zeta(u) = +\infty \tag{3}$$

$$\lim_{u \to 0_+} \sqrt{u}\zeta(u) = 0. \tag{4}$$

We replace the generator L_β by

$$L_\zeta := U\Delta \cdot - \zeta(U)\langle \nabla U, \nabla \cdot \rangle \tag{5}$$

Theorem 0.3. *Assume that*

$$\beta < \min_{n \in [\![N]\!]} \frac{\sum_{i \in [\![m]\!]} \lambda_i(n)}{2\lambda_m(n)} - 1. \tag{6}$$

Whatever the initial condition $X(0) \notin \mathcal{U}$, we have

$$\mathbb{P}\left[\lim_{t \to +\infty} X(t) \text{ exists and belongs to } \mathcal{U}\right] = 0.$$

1. Proof of a weaker version of Theorem 0.1

Here we prove Theorem 0.1 under the stronger assumption

$$\beta > 1 + \sup_{n \in [\![N]\!]} \frac{\sum_{i \in [\![m]\!]} \lambda_i(n)}{2\lambda_1(n)}. \tag{7}$$

Its relaxation to (2) will be shown in the next section.

First let us give a sketch of the proof. For any given global minimum y_n, with $n \in [\![N]\!]$, we can find a small radius $r_n > 0$ such that inside the ball $B(y_n, r_n)$, the evolution of $U(X(t))$ is comparable to a time-changed Bessel process with negative dimension. Since a Bessel process with negative dimension a.s. converges to zero in finite time, X will stay in $B(y_n, r_n)$ forever with positive probability and then converge to y_n, if it happens to belong to an even smaller neighborhood V_n of y_n at some time. Nevertheless, the convergence is not expected to occur in finite time due to the time-change, see Example 1.4 below. It remains to remark that outside $\bigcup_{n \in [\![N]\!]} V_n$, the diffusion X is elliptic, so it will end up entering $\bigcup_{n \in [\![N]\!]} V_n$. Since each times this happens X has a positive chance to converge to a point of $\mathcal{U}$, this event will end up occurring with probability 1. The last assertions of Theorem 0.1 are consequences of the ellipticity of X outside $\mathcal{U}$.

Let us now develop more precisely the above arguments.

We begin by recalling some general facts about the process X, for any fixed $\beta \in \mathbb{R}$. Its law on the set of continuous trajectories from $\mathbb{R}_+$ to M is uniquely determined by the initial distribution of $X(0)$ and by the fact for any smooth function $f : M \to \mathbb{R}$, the process $M^f := (M^f(t))_{t \geq 0}$ defined by

$$\forall \ \ t \geq 0, \quad M^f(t) := f(X(t)) - f(X(0)) - \int_0^t L_\beta[f](X(s))\, ds$$

is a continuous martingale (with respect to the filtration generated by X). Furthermore, its quadratic variation (the notation of which should not to be confused with the Riemannian scalar product) is given by

$$\forall\ t \geq 0, \quad \left\langle M^f \right\rangle_t = 2\int_0^t U(X(s))\|\nabla f(X(s))\|^2\, ds \tag{8}$$

where $\|\cdot\|$ stands for the Riemannian norm.

When M is a flat torus, these observations can be deduced from Itô's formula, asserting that:

$$\begin{aligned} df(X(t)) = \big[U(X(t))\Delta f(X(t)) - \beta\langle \nabla U(X(t)), \nabla f(X(t))\rangle\big]\, dt \\ +\sqrt{2U(X(t))}\,\langle \nabla f(X(t)), dB(t)\rangle \end{aligned}$$

so that

$$\forall\ t \geq 0, \quad M^f(t) = \int_0^t \sqrt{2U(X(s))}\,\langle \nabla f(X(s)), dB(s)\rangle.$$

This formula can be generalized for general compact Riemannian manifolds, nevertheless the above martingale problem point of view is more synthetic and enables to avoid delicate geometric constructions. For extensive developments of the martingale problem approach, we refer to the books of Stroock and Varadhan [SV97] and Ethier and Kurtz [EK86].

Introduce

$$\sigma^f := \inf\Big\{t \geq 0\ :\ U(X(t))\|\nabla f(X(t))\|^2 = 0\Big\}$$

with the convention that $\sigma^f = +\infty$ when the r.h.s. is the empty set. Assume that $U(X(0))\|\nabla f(X(0))\|^2 \neq 0$, so that $\sigma^f > 0$ a.s.

Since for any $t \in [0, \sigma^f)$, we have $U(X(t))\|\nabla f(X(t))\|^2 > 0$, we can consider the time change $(\tau_t^f)_{t\in[0,\varsigma)}$ uniquely defined through

$$\forall\ t \in \big[0, \varsigma^f\big), \quad 2\int_0^{\tau_t^f} U(X(s))\|\nabla f(X(s))\|^2\, ds = t$$

where

$$\varsigma^f := 2\int_0^{\sigma^f} U(X(s))\|\nabla f(X(s))\|^2\, ds.$$

Define the process Y^f via

$$\big(Y^f(t)\big)_{t\in[0,\varsigma^f)} := \Big(f\Big(X\big(\tau_t^f\big)\Big)\Big)_{t\in[0,\varsigma^f)}.$$

Classical time-change theory, see for instance [RY99, Section 1, Chapter 5] of Revuz and Yor, Levy's characterization theorem and (8), enable us to construct a Brownian motion $(W(t))_{t\geq 0}$ (up to a possible enlargement of the underlying probability space), so that for any time $t \in [0, \varsigma)$,

$$dY^f(t) = \frac{1}{2}F^f\big(Y^f(t)\big)\, dt + dW(t)$$

where for any $x \in M$ such that $U(x)\|\nabla f(x)\|^2 \neq 0$,

$$\begin{aligned} F^f(x) &:= \frac{L_\beta[f](x)}{U(x)\|\nabla f(x)\|^2} \\ &= \frac{\Delta f(x)}{\|\nabla f(x)\|^2} - \beta\frac{\langle \nabla U(x), \nabla f(x)\rangle}{U(x)\|\nabla f(x)\|^2}. \end{aligned}$$

Now let us apply these considerations to a particular function f. The first idea is to take $f = U$ (in next section we will see that this is not optimal). To simplify the notations, all the superscripts f are removed in this case. Thus we have

$$\sigma = \inf\{t \geq 0\ :\ X(t) \in \mathcal{C}\}$$

with $\mathcal{C} := \{x \in M\ :\ \nabla U(x) = 0\}$ being the set of critical points of U. Furthermore we consider the process Y given by

$$(Y(t))_{t\in[0,\varsigma)} := (U(X(\tau_t)))_{t\in[0,\varsigma)}$$

with

$$\forall\ t \in [0, \varsigma), \quad 2\int_0^{\tau_t} U(X(s))\|\nabla U(X(s))\|^2\, ds = t$$

and

$$\varsigma := 2\int_0^{\sigma} U(X(s))\|\nabla U(X(s))\|^2\, ds.$$

The evolution of Y is given by

$$dY(t) = \frac{1}{2}F(X(\tau_t))\,dt + dW(t) \tag{9}$$

where $(W(t))_{t\geq 0}$ is a Brownian motion and where for any $x \in M \setminus \mathcal{C}$,

$$F(x) := \frac{\Delta U(x)}{\|\nabla U(x)\|^2} - \frac{\beta}{U(x)}.$$

The process $(U(X(t)))_{t\geq 0}$ is not Markovian in general, nevertheless we will show that it can be conveniently compared to a Bessel process while $X(t)$ is close to an element of $\mathcal{U}$.

Indeed, fix a global minimum y_n, with $n \in [\![N]\!]$. Consider an exponential system of coordinates $(x_1, x_2, \ldots, x_m)$ on a neighborhood $\mathcal{N}_n$ of y_n with $x_1(y_n) = x_2(y_n) = \cdots = x_m(y_n) = 0$ and such that the vectors $(\partial_{x_i})_{i\in[\![m]\!]}$ forms an orthonormal basis of the tangent space at y_n consisting of eigenvectors of the Hessian of U respectively to the eigenvalues $(\lambda_i(n))_{i\in[\![m]\!]}$. Let $\epsilon \in (0,1)$, the value of which will be chosen more precisely below. We can find a small enough radius $r_n > 0$, such that the open ball $B(y_n, r_n)$ is included into $\mathcal{N}_n$ and such that for any $x \in B(y_n, r_n)$, identified with its coordinates $(x_1, x_2, \ldots, x_m)$, we have

$$(1-\epsilon)\sum_{i\in[\![m]\!]}\lambda_i(n) \leq \Delta U(x) \leq (1+\epsilon)\sum_{i\in[\![m]\!]}\lambda_i(n) \tag{10}$$

$$(1-\epsilon)\sum_{i\in[\![m]\!]}\lambda_i^2(n)x_i^2 \leq \|\nabla U\|^2(x) \leq (1+\epsilon)\sum_{i\in[\![m]\!]}\lambda_i^2(n)x_i^2 \tag{11}$$

$$\frac{1}{2}(1-\epsilon)\sum_{i\in[\![m]\!]}\lambda_i(n)x_i^2 \leq U(x) \leq \frac{1}{2}(1+\epsilon)\sum_{i\in[\![m]\!]}\lambda_i(n)x_i^2. \tag{12}$$

Indeed, the first and second estimates are obtained through the expressions of the Laplace–Beltrami and gradient operators in terms of the metric, taking into account that the first order expansion of the metric in a exponential (or geodesic) chart is constant, see for instance [CLN06, Section 10.1, p. 58] of Chow, Lu and Ni.

Remark 1.1. For $x \in M \setminus \mathcal{U}$, we have $\nabla\sqrt{U} = \nabla U/(2\sqrt{U})$ and the above bounds enable to see the norm of this vector is bounded on $B(y_n, r_n) \setminus \{y_n\}$, for all $n \in [\![N]\!]$. Since $\|\nabla\sqrt{U}\|$ is clearly bounded on $M \setminus \bigcup_{n\in[\![N]\!]} B(y_n, r_n)$, we deduce that $\sqrt{U}$ is Lipschitz, as announced in the introduction.

In particular, y_n is the unique critical point of U in $B(y_n, r_n)$. Assume that $X(0) \in V_n$ with

$$V_n := \{x \in B(y_n, r_n) \setminus \{y_n\} \,:\, U(x) < u_n/2\} \tag{13}$$

with

$$u_n := \min_{z\,\in\,\partial B(y_n, r_n)} U(z) \tag{14}$$

where $\partial B(y_n, r_n)$ is the boundary of $B(y_n, r_n)$. Denote

$$\sigma_n := \inf\{t \geq 0 \,:\, X(t) = y_n \quad \text{or} \quad X(t) \notin B(y_n, r_n)\} \;\leq\; \sigma \tag{15}$$

(as usual, $\sigma_n = +\infty$ when the r.h.s. is the empty set), and

$$\varsigma_n := 2\int_0^{\sigma_n} U(X(s))\|\nabla U(X(s))\|^2\,ds.$$

Replacing ς by $\varsigma_n \leq \varsigma$ in the above considerations enables to investigate the process X while it stays in $B(y_n, r_n/2) \setminus \{y_n\}$. We are led to study $(Y(t))_{t\,\in\,[0,\varsigma_n]}$. The following result is important in this respect.

Lemma 1.2. *We have for any $x \in B(y_n, r_n) \setminus \{y_n\}$,*

$$\left(\frac{(1-\epsilon)^2\sum_{i\in[\![m]\!]}\lambda_i(n)}{2(1+\epsilon)\lambda_m(n)} - \beta\right)\frac{1}{U(x)} \;\leq\; F(x) \;\leq\; \left(\frac{(1+\epsilon)^2\sum_{i\in[\![m]\!]}\lambda_i(n)}{2(1-\epsilon)\lambda_1(n)} - \beta\right)\frac{1}{U(x)}.$$

Proof. For any $x \in B(y_n, r_n) \setminus \{y_n\}$, we have

$$\frac{(1-\epsilon)\sum_{i\in[\![m]\!]}\lambda_i(n)x_i^2}{(1+\epsilon)\sum_{i\in[\![m]\!]}\lambda_i^2(n)x_i^2} \;\leq\; \frac{2U(x)}{\|\nabla U(x)\|^2} \;\leq\; \frac{(1+\epsilon)\sum_{i\in[\![m]\!]}\lambda_i(n)x_i^2}{(1-\epsilon)\sum_{i\in[\![m]\!]}\lambda_i^2(n)x_i^2}$$

implying

$$\frac{(1-\epsilon)}{(1+\epsilon)\lambda_m(n)} \;\leq\; \frac{2U(x)}{\|\nabla U(x)\|^2} \;\leq\; \frac{(1+\epsilon)}{(1-\epsilon)\lambda_1(n)}$$

since for any real numbers x_i, for $i \in [\![m]\!]$, not all of them vanishing, we have

$$\frac{1}{\lambda_m(n)} \;\le\; \frac{\sum_{i\in[\![m]\!]}\lambda_i(n)x_i^2}{\sum_{i\in[\![m]\!]}\lambda_i^2(n)x_i^2} \;\le\; \frac{1}{\lambda_1(n)}.$$

The announced estimate follows, by writing

$$\forall\ x\in B(y_n,r_n)\setminus\{y_n\},\quad F(x)=\left(\Delta U(x)\frac{U(x)}{\|\nabla U(x)\|^2}-\beta\right)\frac{1}{U(x)}. \qquad \square$$

Recall that β has been chosen so that

$$\beta > 1+\sup_{n\in[\![N]\!]}\frac{\sum_{i\in[\![m]\!]}\lambda_i(n)}{2\lambda_1(n)}$$

so we can choose $\epsilon>0$ so that

$$\delta := \frac{(1+\epsilon)^2\sum_{i\in[\![m]\!]}\lambda_i(n)}{2(1-\epsilon)\lambda_1(n)}+1-\beta$$

is negative.

We deduce from (9) that

$$\forall\ t\in[0,\varsigma_n),\quad dY(t)\le dW(t)+\frac{\delta-1}{2Y(t)}\,dt. \tag{16}$$

This inequality leads us to consider $\widetilde{Y}:=(\widetilde{Y}(t))_{t\ge 0}$ solution of the stochastic differential equation

$$\forall\ t\ge 0,\quad d\widetilde{Y}(t)=dW(t)+\frac{\delta-1}{2\widetilde{Y}(t)}\,dt \tag{17}$$

starting with $\widetilde{Y}(0)=u_n/2$ (defined in (14)).

The process $\widetilde{Y}$ is a Bessel process with negative dimension $\delta<0$, for a recent account, see e.g. Le Gall [LG15]. It hits 0 in (a.s.) finite time and stays at 0 afterward, the strength of the drift not allowing it to escape from 0.

Define

$$\begin{aligned}\theta &:= \inf\Big\{t\ge 0\ :\ \widetilde{Y}(t)=0\Big\}\\ \widetilde{\theta} &:= \inf\Big\{t\ge 0\ :\ \widetilde{Y}(t)=u_n\Big\}\\ p_n &:= \mathbb{P}[\theta<\widetilde{\theta}].\end{aligned} \tag{18}$$

Since $\widetilde{Y}$ is a Markov process and that $\lim_{t\to+\infty}\widetilde{Y}(t)=0$ a.s., we have $p_n>0$.

By comparison, we get

Lemma 1.3. *Assume that $X(0)\in V_n$ defined in* (13)*. Then we have*

$$\mathbb{P}\left[\forall\ t\ge 0,\ X(t)\in B(y_n,r_n)\quad \textit{and}\quad \lim_{t\to+\infty}X(t)=y_n\right]\ge p_n.$$

Proof. For $\epsilon\in(0,Y(0))$, define

$$\theta_\epsilon := \inf\{t\ge 0\ :\ \widetilde{Y}(t)=\epsilon\}.$$

Applying [RY99, Theorem 3.7] of Revuz and Yor, we get that

$$\forall\ t\in[0,\theta_\epsilon\wedge\widetilde{\theta}),\quad Y(t)\le\widetilde{Y}(t).$$

Indeed, this a consequence of $Y(0)\le u_n/2=\widetilde{Y}(0)$ as well as of the comparison between (16) and (17). Note that in the proof of [RY99, Theorem (3.7)] of Revuz and Yor, we need $\widetilde{Y}$ to be Markovian and that its drift is Lipschitz, which is true as long as it belongs to the segment $[\epsilon,u_n]$. But Y is not required to be Markovian, it is sufficient that its drift is adapted.

Letting ϵ go to zero, we deduce

$$\forall\ t\in[0,\theta\wedge\widetilde{\theta}),\quad Y(t)\le\widetilde{Y}(t)$$

(one would have remarked that $Y(t)$ remains strictly below u_n and thus $X(\tau_t)$ remains in $B(y_n,r_n)$ as long as $t<\widetilde{\theta}$).

It follows that on the event $\{\theta<\widetilde{\theta}\}$, we have that Y hits 0 in finite time. Equivalently, $X(\tau_t)$ has to hit y_n in finite time, since $X(\tau_t)$ has also to remain in $B(y_n,r_n)$ on $\{\theta<\widetilde{\theta}\}$ and y_n is the unique point of this ball where U vanishes. Recall that if X hits y_n, then it has to stay there forever. Thus on the event $\{\theta<\widetilde{\theta}\}$, $X(t)$ always stays in $B(y_n,r_n)$ and $\lim_{t\to+\infty}X(t)=y_n$. It leads to the announced result. $\square$

Due to the time change, we cannot deduce the convergence of X to y_n in finite time from the corresponding convergence of $(X_{\tau_t})_{t\geq 0}$ (when it occurs). On the contrary, we believe that X never converges in finite time, as suggested by the following caricatural example.

Example 1.4. Let us momentarily leave the compact setting and consider the function $U : \mathbb{R} \ni x \mapsto x^2/2$. The corresponding diffusion X on $\mathbb{R}$ given by (1) solves the stochastic differential equation

$$\forall\ t \geq 0, \quad dX(t) = -\beta X(t)dt + |X(t)|dB(t).$$

Assume that $X(0) > 0$ and consider

$$\sigma := \inf\{t \geq 0 \,:\, X(t) = 0\}.$$

While $t \in [0, \sigma)$, we have $X(t) > 0$, so that

$$\forall\ t \in [0, \sigma), \quad dX(t) = -\beta X(t)dt + X(t)dB(t).$$

It leads us to introduce $(\widetilde{X}(t))_{t\geq 0} := (e^{\beta t}X(t))_{t\geq 0}$, satisfying

$$\forall\ t \in [0, \sigma), \quad d\widetilde{X}(t) = \widetilde{X}(t)dB(t)$$

whose solution is well-known to be the exponential martingale

$$\forall\ t \in [0, \sigma), \quad \widetilde{X}(t) = \widetilde{X}(0)\exp(B(t) - t/2).$$

It follows that

$$\forall\ t \in [0, \sigma), \quad X(t) = X(0)\exp(B(t) - (1 + 2\beta)t/2).$$

By consequence we have $\sigma = +\infty$ and the process X does not hit the global minima of U in finite time. Note also that $\lim_{t\to+\infty} X(t) = 0$ as soon as $\beta > -1/2$ in accordance with the observation following Corollary 0.2.

This example can be transferred to the compact space $\mathbb{R}/(2\pi\mathbb{Z})$ in the following way: consider on $\mathbb{R}/(2\pi\mathbb{Z})$ a Morse function $\widetilde{U}$ coinciding with the above U on $[-\pi/2, \pi/2]$ and such that 0 is a global minimum of $\widetilde{U}$. Let $\widetilde{X}$ be the diffusion evolving as (1), but with U replaced by $\widetilde{U}$. Assume that $\widetilde{X}(0) = \pi/4$ and by contradiction that the corresponding hitting time $\widetilde{\sigma}$ of 0 is finite with positive probability. Taking into account the Markov property, the process $\widetilde{X}$ then stays inside $(-\pi/2, \pi/2)$ and converges to 0 in finite time with a positive probability $p > 0$ (since if starting from $\pi/4$, $\widetilde{X}$ always hits $\pi/2$ before 0, it cannot converge to 0). It follows that if X also starts from $\pi/4$ and uses the same driving Brownian motion, then it coincides with $\widetilde{X}$ for all times with probability p. This implies that X converges to zero in finite time with positive probability, a contradiction.

The fact that the diffusion given by (1) does not hit $\mathcal{U}$ in finite time with positive probability is probably true under the Morse assumption of this paper.

The end of the proof of Theorem 0.1 follows the pattern sketched at the beginning of this section. More precisely, define

$$\begin{aligned}
\widetilde{A} &:= M \setminus \bigcup_{n\,\in\,[\![N]\!]} B(y_n, r_n)\\
\widehat{A} &:= \bigcup_{n\,\in\,[\![N]\!]} V_n\\
p &:= \min\{p_n \,:\, n \in [\![N]\!]\} \ > \ 0.
\end{aligned}$$

Whatever the initial distribution of $X(0)$, consider the sequences of stopping times $(\widetilde{\theta}_k)_{k\geq 0}$ and $(\widehat{\theta}_k)_{k\geq 0}$ defined iteratively from $\widetilde{\theta}_0 = 0$ via

$$\begin{cases}
\widehat{\theta}_k := \inf\Big\{t \geq \widetilde{\theta}_k \,:\, X(t) \in \widehat{A}\Big\}\\
\widetilde{\theta}_{k+1} := \inf\Big\{t \geq \widehat{\theta}_k \,:\, X(t) \in \widetilde{A}\Big\}
\end{cases} \tag{19}$$

(with the usual convention that the infimum of the empty set is $+\infty$).

By the strong ellipticity of X on the compact set $M \setminus \widehat{A}$, for any $k \in \mathbb{N}$ such that $\widetilde{\theta}_k < +\infty$, we have a.s. $\widehat{\theta}_k < +\infty$. On the contrary, we deduce from Lemma 1.3 and from the strong Markov property that for any $k \in \mathbb{N}$ such that $\widehat{\theta}_k < +\infty$, we end up with $\widetilde{\theta}_{k+1} = +\infty$ with probability p at least. It follows that for any $k \in \mathbb{Z}_+$,

$$\mathbb{P}\Big[\widetilde{\theta}_{k+1} < +\infty \,\Big|\, \widetilde{\theta}_k < +\infty\Big] \leq (1 - p)$$

so by iteration we get

$$\forall\ k \in \mathbb{Z}^+, \quad \mathbb{P}\Big[\widetilde{\theta}_k < +\infty\Big] \leq (1 - p)^k$$

and finally

$$\mathbb{P}\Big[\forall\ \ k\in\mathbb{Z}_+,\,\widetilde{\theta}_k<+\infty\Big]=0.$$

The fact that a.s. there exists a random $k\in\mathbb{Z}_+$ such that $\widetilde{\theta}_k=+\infty$ ends the proof of the first statement of Theorem 0.1.

Concerning its second statement, note that when $X(0)\in V_n$, for some $n\in[\![N]\!]$, there is a positive probability that X exits $B(y_n,r_n)\setminus\{y_n\}$ via the boundary $\partial B(y_n,r_n)$.

When $m\geq 2$, assume that the $r_n>0$, for $n\in[\![N]\!]$, have been furthermore chosen so small so that $M\setminus\bigsqcup_{n\in[\![N]\!]}V_n$ is connected and contains $\bigsqcup_{n\in[\![N]\!]}\partial B(y_n,r_n)$ (the $\sqcup$ meaning that it is a union of disjoint sets). It follows that if $X(0)\in\bigsqcup_{n\in[\![N]\!]}\partial B(y_n,r_n)$, then by ellipticity of L_β on the connected set $M\setminus\bigsqcup_{n\in[\![N]\!]}V_n$, for any $n\in[\![N]\!]$, we have $\mathbb{P}[X(\widehat{\theta}_1)\in B(y_n,r_n)]>0$ for any given $n\in[\![N]\!]$. Taking into account that $\mathbb{P}[\widetilde{\theta}_2=+\infty|X(\widehat{\theta}_1)\in B(y_n,r_n)]>0$, we deduce that $\mathbb{P}[X(\infty)=y_n]>0$.

When $m=1$, a similar reasoning leads to the desired result, ending the proof of Theorem 0.1 under (7).

2. Extensions

After ending the proof of Theorem 0.1, we show Theorem 0.3 and next we present the extension to generators of the form (5) and discuss the Morse assumption.

2.1. **Under Assumption** (2)**.** We end the proof of Theorem 0.1. The overall approach is the same but instead of considering $f=U$, we take $f=U^a$, with $a\in(0,1)$. This exponent being fixed, we remove all the superscripts f from the notations, as in the previous section when f was equal to U. An immediate drawback with respect to the latter case is that a priori U^a is not smooth. But it is not difficult to go around this problem, since U^a is smooth on $M\setminus\mathcal{U}$ and we are only considering our process up to the time it may reach $\mathcal{U}$.

More precisely, we proceed as in the previous section, assuming that $X(0)\in V_n\setminus\{y_n\}$ for some $n\in[\![N]\!]$. Instead of considering σ_n given in (15), we introduce for any $\epsilon\in(0,U(X(0)))$ the stopping time

$$\sigma_{n,\epsilon}:=\inf\{t\geq 0\,:\,X(t)\notin A_{n,\epsilon}\}$$

with

$$A_{n,\epsilon}:=\{x\in B(y_n,r_n)\,:\,U(x)\geq\epsilon\}.$$

Remark that

$$\lim_{\epsilon\to 0_+}\sigma_{n,\epsilon}=\sigma_n.$$

The advantage of $A_{n,\epsilon}$ is that there exist smooth functions f on M coinciding with U^a on $A_{n,\epsilon}$. The considerations at the beginning of Section 1 can be applied to such a function f, up to the time $\sigma_{n,\epsilon}$. By letting ϵ go to 0_+, we get the following result, analogous to (9).

Consider the process Y given by

$$(Y(t))_{t\in[0,\varsigma_n)}:=(U(X(\tau_t)))_{t\in[0,\varsigma_n)}$$

with

$$\forall\ \ t\in[0,\varsigma_n),\quad 2\int_0^{\tau_t}U(X(s))\|\nabla U^a(X(s))\|^2\,ds=t$$

and

$$\varsigma_n:=2\int_0^{\sigma_n}U(X(s))\|\nabla U^a(X(s))\|^2\,ds.$$

The evolution of Y is given by

$$dY(t)=\frac{1}{2}F(X(\tau_t))\,dt+dW(t)$$

where $(W(t))_{t\geq 0}$ is a Brownian motion and for any $x\in M\setminus\mathcal{C}$,

$$\begin{aligned}F(x)&:=\frac{\Delta U^a(x)}{\|\nabla U^a(x)\|^2}-\beta\frac{\langle\nabla U(x),\nabla U^a(x)\rangle}{U(x)\|\nabla U^a(x)\|^2}\\&=\frac{\Delta U^a(x)}{a^2U^{2a-2}(x)\|\nabla U\|^2}-\frac{\beta}{aU^a(x)}.\end{aligned}$$

We compute that outside $\mathcal{C}$,

$$\Delta U^a=aU^{a-1}\Delta U+a(a-1)U^{a-2}\|\nabla U\|^2$$

so that outside $\mathcal{C}$,

$$F = \frac{1}{a}\left(\frac{U\Delta U}{\|\nabla U\|^2} + a - 1 - \beta\right)\frac{1}{U^a}.$$

From Lemma 1.2, we get that a comparison is possible with a Bessel process of asymptotic dimension (i.e. when we let ϵ go to zero in Lemma 1.2)

$$\delta_a = 2 + \frac{1}{a}\left(\frac{\sum_{i\in[\![m]\!]}\lambda_i(n)}{2\lambda_1(n)} - 1 - \beta\right).$$

Thus under (2), by choosing $a > 0$ sufficiently small, we can get $\delta_a < 0$, ending the proof of Theorem 0.1.

Remark 2.1. Without resorting to U^a, Condition (2) can also be obtained through the following observation. Coming back to the argument of the previous section, what is important for our purpose is that the Bessel process (17) can hit 0 in finite time, since this implies the same behavior for the process Y. Indeed, once X has hit y_n, it cannot escape from it, so it closes our time-horizon (this is even more true when X does not hit y_n in finite time as suggested by Remark 1.4). It remains to recall that a Bessel process of dimension δ hits 0 in finite time if and only if $\delta < 2$. This leads to Condition (2). But this argument conceals there is a negative (even as negative as we want) dimension Bessel process behind the scene.

2.2. **Proof of Theorem 0.3.** Up to now we have not taken into account the lower bound in Lemma 1.2. It can be used to prove there is no convergence toward $\mathcal{U}$. More precisely, assume that (6) holds and let us show we cannot have $\lim_{t\to+\infty} X(t) \in \mathcal{U}$ (except on a $\mathbb{P}$-negligible set).

Indeed, the lower bound in Lemma 1.2 enables to make a reverse comparison between Y and the Bessel process $\widetilde{Y}$ of (asymptotic) dimension

$$\delta := \min_{n\in[\![N]\!]} \frac{\sum_{i\in[\![m]\!]}\lambda_i(n)}{2\lambda_m(n)} + 1 - \beta$$

which is (strictly) larger than 2 under (6). Such a Bessel process diverges to $+\infty$ in large time without hitting 0 (starting from a positive value). It implies that the time σ_n always satisfies that $X(\sigma_n)$ belongs to the boundary of the ball $B(y_n, r_n)$. If we come back to (19), we get that for any $k \in \mathbb{Z}_+$, $\widetilde{\theta}_k < +\infty$. In particular the set $M \setminus \bigcup_{n\in[\![N]\!]} B(y_n, r_n)$ is visited again after any given time: the convergence to $\mathcal{U}$ is thus forbidden.

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